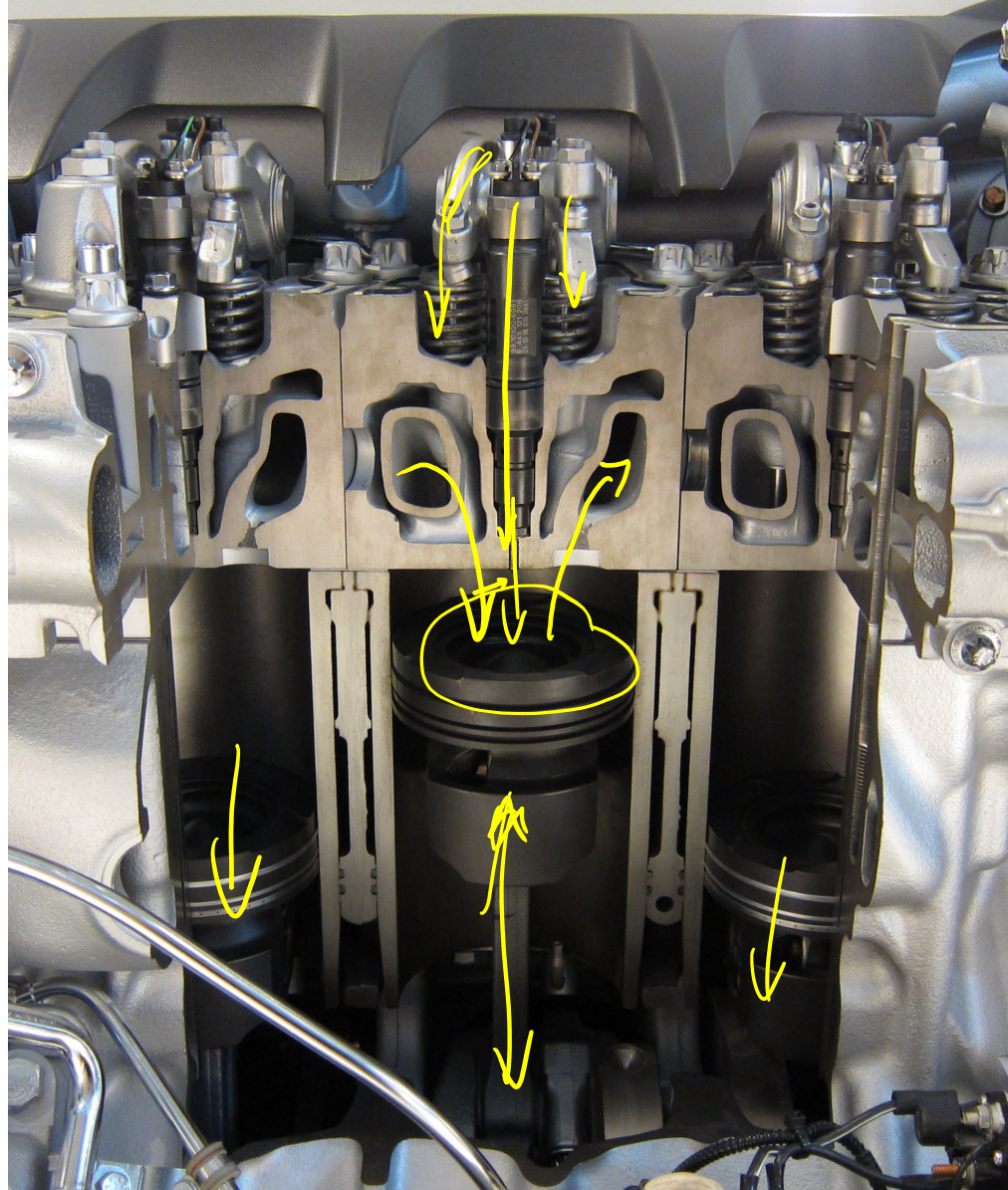


ME 300:
Engineering Thermodynamics

Example 1:
Ideal Gas Equation
of State: P - v - T

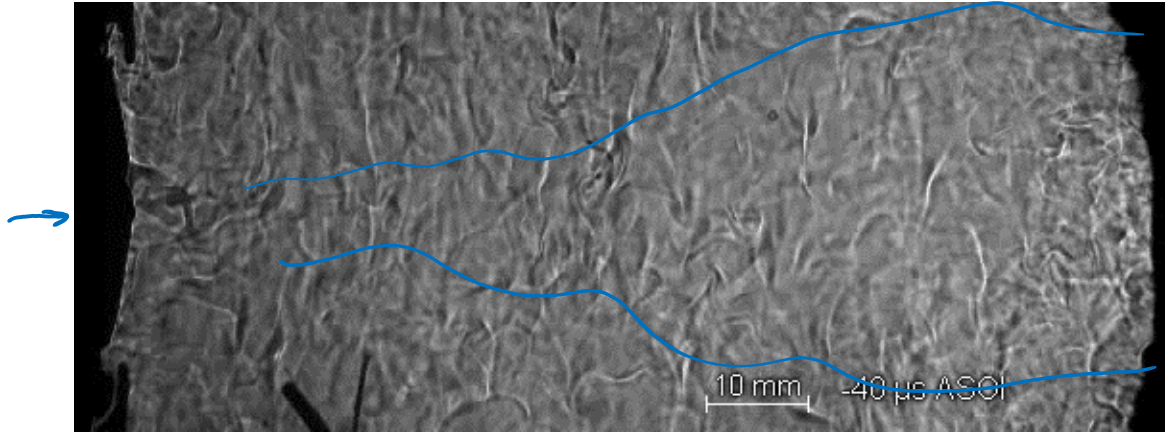
Diesel fuel injection



Diesel fuel injection

Condition 1 *air*

- $T = 900 \text{ K}$
- $P = 5.92 \text{ Mpa}$ *$\sim 60 \text{ atm}$*
- $\rho = 22.8 \text{ kg/m}^3$

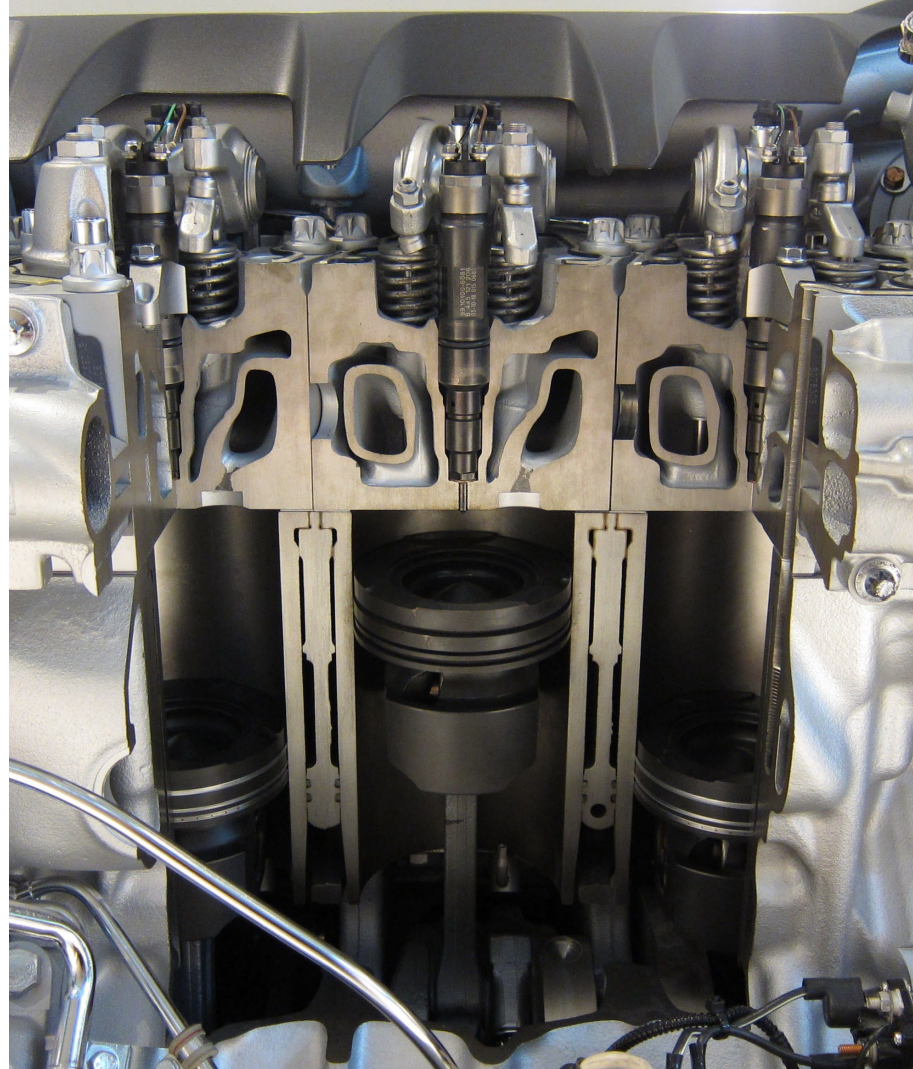


Condition 2

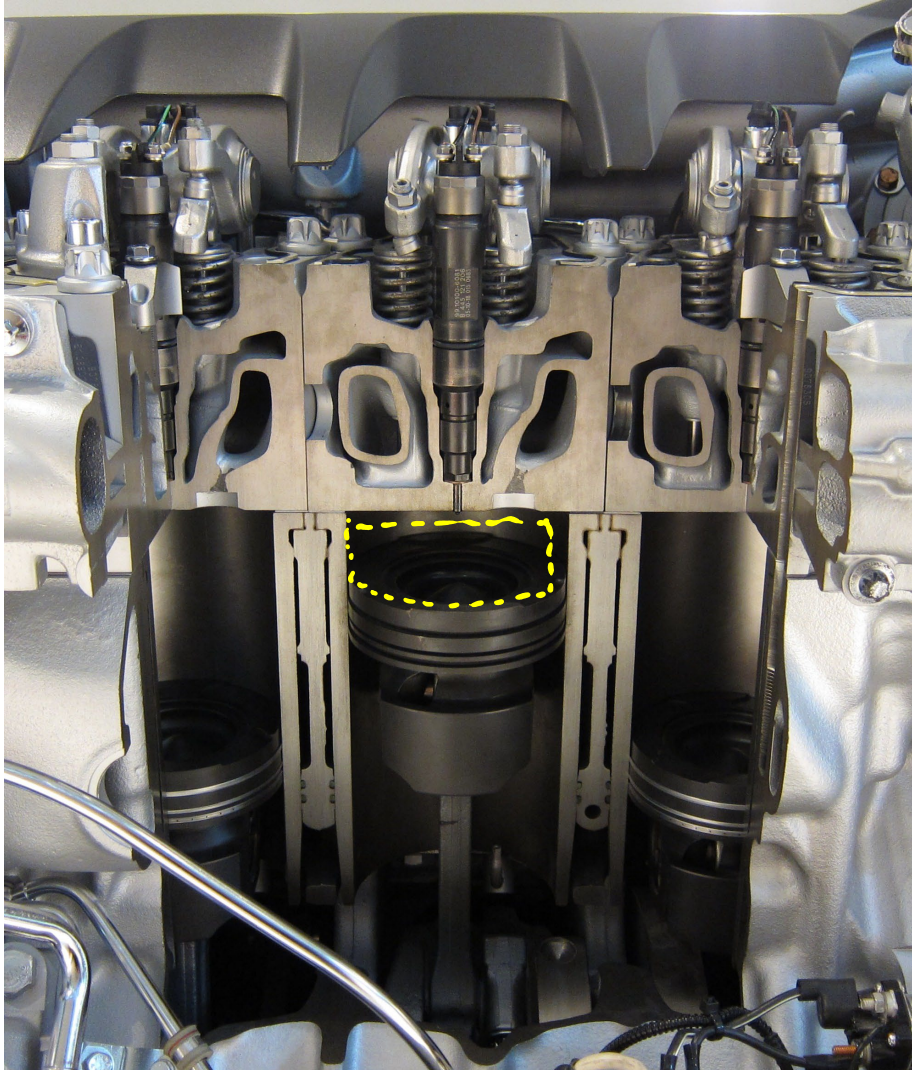
- $T = 900 \text{ K}$
- $P = 4.03 \text{ Mpa}$
- $\rho = 15.2 \text{ kg/m}^3$



How would we design a compression from $T_1 = 289 \text{ K}$ to achieve these conditions?



What control volume should we use?



=> only include the stuff you
care about → solve for



Which of the following assumptions should we make?

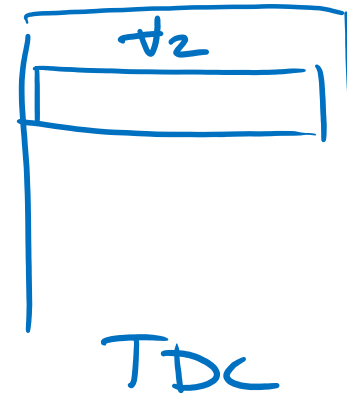
air

- a) Pure substance ✓
- b) Simple compressible substance ✓
- c) Continuum ✓
- d) Ideal gas ✓
- e) All of the above

Compression ratio

Displacement volume: $V_1 - V_2$

Compression ratio: $\frac{V_1}{V_2} = 11.2$



Inlet pressure

$$\frac{V_1}{V_2} = 11.2 \checkmark$$

$$T_1 = 289 \text{ K} \checkmark$$

$$T_2 = 900 \text{ K}$$

$$P_2 = 5.92 \text{ MPa}$$

$$P_1 = ?$$

$$\Rightarrow M = \text{const.}$$

$$R = \text{const.}$$

$$PV = \overbrace{MRT}^{\text{const } 1,2}$$
$$M_1 R = M_2 R \rightarrow$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\sqrt{\frac{V_1}{V_2}} = \frac{P_2 T_1}{P_1 T_2}$$

$$P_1 = \frac{P_2 T_1}{T_2} \cdot \frac{1}{V_1/V_2}$$

$$= \frac{(5.92 \times 10^6)(289)}{(900)} \cdot \frac{1}{11.2}$$

$$P_1 = 169700 \text{ Pa}$$